

Subject Description Form

Subject Code	EE505
Subject Title	Power System Control and Operation
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	1. To introduce the concept of modern power system control & operation to students; 2. To integrate theory and practical knowledge of power system control & operation; 3. To understand the working principle of power system control and operation; 4. To apply the theory in power system control & operation; and 5. To understand the industrial practice and tools used in power system control and operations
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Ability to analyse power system security control & operation; b. Ability to analyse interconnected power system interchange and economic operation. c. Ability to analyse power system computer control and applications; d. Understand the functionalities and able to use to appropriate level of competence of selected specialty software for power system control and operation purpose; e. To be aware of new technologies development trends and environmental impacts of modern power system control and operation techniques; and f. Ability to write technical reports and present the findings through individual effort as well as team work
Subject Synopsis/ Indicative Syllabus	1. Power system operational security and dispatch: Power system security concepts. Contingency analysis. Static and dynamic security. States of operation. Prevention of blackouts. Power system state estimation concepts. Application of state estimation. 2. Unit commitment and economic dispatch: Priority lists. Methodologies for large system economic dispatch and unit commitment. Programming methods. 3. Frequency and voltage control: Frequency and voltage control concepts. Control loops and analysis. Automatic generation control (AGC) concepts, methodology and implementation. 4. Interconnected systems operation: System interconnection merits and problems. Economic interchange and control. Multi-area operation. 5. Energy management and real-time control: Energy management systems. Software systems. Computer hardware resources and configurations. Data management. Communication and distributed computing. Load forecasting. Contingency and security assessment. System restoration and emergency control concepts. Case Study: 1. Local system control centre arrangement. 2. Case study of past system blackout in overseas countries. 3. AGC and voltage control case studies. 4. Power system developments in HK and China as well as overseas countries. 5. Applications of computer technology in power system control and monitoring

Teaching/Learning Methodology	Lectures and tutorials are the primary means of conveying the basic concepts and theories. Experiences on real world cases and associated analysis are given through case studies, in which the students are expected to power system control and operation problems with real-life constraints and to attain pragmatic solutions with critical and analytical thinking. Guest lecture / industrial seminars will be given to provide hands-on experience and knowledge on this subject from industry practice. Mini-project is designed to supplement the lecturing materials so that the students are encouraged to take extra readings and practice specialty software tools for power system operation and control.																																																												
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		a	b	c	d	e	f																																																						
	Lectures	√	√	√	√																																																								
	Tutorials	√	√	√	√																																																								
	Report	√	√	√	√	√	√																																																						
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><td rowspan="2">Specific assessment methods/tasks</td><td rowspan="2">% weighting</td><td colspan="6">Intended subject learning outcomes to be assessed</td></tr><tr><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td><td>f</td></tr><tr><td>1. Exam</td><td>60%</td><td>√</td><td>√</td><td>√</td><td></td><td>√</td><td></td></tr><tr><td>2. Class test</td><td>18%</td><td>√</td><td>√</td><td>√</td><td></td><td>√</td><td></td></tr><tr><td>3. Mini-project & report</td><td>12%</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>4. Essay Assignment</td><td>10%</td><td>√</td><td></td><td></td><td></td><td>√</td><td>√</td></tr><tr><td>Total</td><td>100%</td><td colspan="6"></td></tr></table> The assessment methods include an examination, a class test, and written assignment in the form of mini-project report. The examination and class test assess the technical competence of students in power system analysis methods and methods of power system operation and control. The written reports assess the students’ ability to apply the theories learned in class to practical project, and to communicate in written form.							Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed						a	b	c	d	e	f	1. Exam	60%	√	√	√		√		2. Class test	18%	√	√	√		√		3. Mini-project & report	12%	√	√	√	√	√	√	4. Essay Assignment	10%	√				√	√	Total	100%						
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Student Study Effort Expected	Class contact:																																																												
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	▪ Essay assignment/Self-study			51 Hrs.																																																									
	Total student study effort			105 Hrs.																																																									
Reading List and References	Reference books: 1. W.D. Stevenson, Elements of Power System Analysis, McGraw Hill 2. Wood & Wollenberg, Power Generation, Operation and Control, J. Wiley. 3. Weedy and Cory, Electric Power Systems, 4 th Edition, Wiley 4. Grainger & Stevenson, Power System Analysis, McGraw Hill 5. H. Saadat, Power System Analysis, McGraw Hill 6. Antonio Gomez-Exposito, Antonio J. Conejo, and Claudio Canizares, Electric Energy Systems: Analysis and Operation, CRC Press, 2009																																																												